



Started on	Monday, 10 August 2026, 10:25 AM
State	Finished
Completed on	Monday, 10 August 2026, 10:31 AM
Time taken	6 mins 26 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main()
3  {
4      int n;
5      int i = 1;
6      int s = 1;
7      int count = 0;
8      scanf("%d", &n);
9      count+=2;
10     while (s <= n)
11     {
12         count++;
13         i++;
14         count++;
15         s += i;
16         count++;
17     }
18     count++;
19     printf("%d", count);
20     return 0;
21 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Monday, 10 August 2026, 10:31 AM
State	Finished
Completed on	Monday, 10 August 2026, 10:38 AM
Time taken	6 mins 45 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2  int main()
3  {
4      int n, count = 1;
5      scanf("%d", &n);
6      if (n == 1)
7      {
8          count++;
9      }
10     else
11     {
12         int i, j;
13         for (i = 1; i <= n; i++)
14         {
15             count++; // i <= n
16             for (j = 1; j <= n; j++)
17             {
18                 count++;
19                 count++;
20                 count++;
21                 break;
22             }
23             count++;
24         }
25         count++;
26     }
27     printf("%d", count);
28     return 0;
```

```
29 | }  
30 |  
31 |
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Monday, 10 August 2026, 10:42 AM
State	Finished
Completed on	Monday, 10 August 2026, 10:46 AM
Time taken	3 mins 44 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2  int main()
3  {
4      int n, count = 0;
5      scanf("%d", &n);
6      int i;
7      for (i = 1; i <= n; ++i)
8      {
9          count++;
10         count++;
11         if (n % i == 0)
12         {
13             count++;
14         }
15     }
16 }
17 count++;
18 printf("%d", count);
19 return 0;
20 }
21
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! 

Correct

Marks for this submission: 1.00/1.00.

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Started on	Monday, 10 August 2026, 10:46 AM
State	Finished
Completed on	Monday, 10 August 2026, 10:57 AM
Time taken	11 mins 19 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1

Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n

Output:
Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2  int main()
3  {
4      int n,i,j,k;
5      int c=2;
6      scanf("%d",&n);
7      for(i=n/2;i<n;i++){
8          c+=2;
9          for(j=1;j<n;j=2*j){
10             c+=2;
11             for(k=1;k<n;k=k*2)
12                 {
13                     c+=2;
14                 }
15             }
16         }
17     }
18     printf("%d",c);
19     return 0;
20 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Monday, 10 August 2026, 10:58 AM
State	Finished
Completed on	Monday, 10 August 2026, 11:00 AM
Time taken	2 mins 2 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1

Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n

Output:
Print the value of the counter variable

Answer:

```
1 | #include<stdio.h>
2 | int main()
3 | {
4 |     int n, rev = 0, remainder;
5 |     int count = 2;
6 |     scanf("%d", &n);
7 |     while (n != 0)
8 |     {
9 |         count++;
10 |        remainder = n % 10;
11 |        count++;
12 |        rev = rev * 10 + remainder;
13 |        count++;
14 |        n /= 10;
15 |        count++;
16 |    }
17 |    count++;
18 |    printf("%d", count);
19 |    return 0;
20 | }
21 |
22 |
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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